20 From Legacy Data to Spatial Models: A Reproducible FLOSS Workflow for Early Medieval Ceramic Imports in Northern Italy (6th–8th Centuries AD)

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The rise of FLOSS technologies and advanced spatial statistics in archaeology has reopened the epistemological debate on the use of legacy data. Unlike datasets natively designed for geostatistical analysis, published records constitute a heterogeneous body of information affected by structural issues: lack of standardized quantification criteria, partial data reporting, taxonomic instability of typo-chronologies, and spatial bias caused by uneven research intensity. Such limitations often hinder the use of certain artifact classes, such as pottery, as proxy indicators for reconstructing large-scale historical-economic dynamics.

This paper presents an open, reproducible workflow using FLOSS tools (SpatiaLite/QGIS and R) to normalize and analyze legacy data on Mediterranean ceramic imports in Northern Italy (6th– 8th c. AD). Based on a binary formalization (0/1) at the context level to overcome quantitative comparability issues between sites, we illustrate a spatial analytical pipeline for 15 ceramic classes integrating weighted Kernel Density Estimation (KDE), Hierarchical Clustering assessed using Silhouette scores, Multinomial Logistic Regression, and Network Analysis (Delaunay and Relative Neighbour Graphs).

This workflow demonstrates how advanced tools enable moving from a traditional archaeological distribution map to a "map-model" capable of managing the "MEAN" (Miscellaneous Exceptional Arbitrary Nonconformist) nature of legacy data (Huvila 2019). The results provide a spatial model of early medieval economic dynamics in Northern Italy, highlighting a shift from widespread distribution in the 6th century to marked spatial polarization in the 7th–8th centuries. Scripting in R ensures granular, transparent parameter control, offering an open, reproducible methodology capable of expanding the dataset with further archaeological evidence.